SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – EXPERIMENTS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO3: Experiments (BL4, BL5)	**Assessment:**	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	V.LASYA	**Reg. No.:**	192524186
Section / Batch:	B.TECH AI&DS	**Date:**	12-08-2026

Instructions to Students

- Write algorithm and execute code both experiments.
- Each questions 50 Marks. Total: 100 Marks.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

Question Type	Description	Questions	Total Marks
Experiments	Design and Code for Divide and Conquer Algorithms: Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication	Q1 – Q2	100

SECTION A – Experiments

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DAA PROGRAMMING

Vandana Lasya
192524186RA

CO3-AT1-Experiments

Back

Language: Python | C | C++ | Java

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer

Q1 **Divide and Conquer** 50 marks

Evaluate the scalability of Merge Sort by applying it to very large datasets (e.g., 10? to 10? elements). Record execution time and plot performance graphs. Analyze the trend and interpret how the algorithm scales with input size. Justify its suitability for large-scale applications

Your Code

```
def merge(a):
    if len(a)>1:
        m=len(a)//2
        L=a[:m]
        R=a[m:]
        merge(L)
        merge(R)
        i=j=k=0
        while i<len(L)and j<len(R):
          if L[i]<R[j]:
             a[k]=L[i]
             i+=1
          else:
             a[k]=R[j]
             j+=1
          k+=1
        a[k:]=L[i:]+R[j:]
n=int(input())
a=list(range(n,0,-1))
merge(a)

print("sorted successfully")
```

Run | Save | Submit

Input

```
1000
```

Output

```
sorted successfully
```

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Language: Python | C | C++ | Java

QUESTIONS

Q1
Divide and Conquer
50 marks

Q2
Divide and Conquer
50 marks

Q2 **Divide and Conquer** 50 marks

Analyze the behavior of Binary Search in terms of cache performance by executing it on large datasets stored in memory. Record execution time for different access patterns. Interpret how memory locality influences performance. Justify the role of data organization in improving search efficiency.

Your Code

```python
import time
def binary(a,x):
  l=0
  r=len(a)-1
  while l<=r:
    m=(l+r)//2
    if a[m]==x:
       return m
    if a[m]<x:
       l=m+1
    else:
      r=m-1
  return-1
n=int(input())
a=list(range(n))
s=time.time()
binary(a,n//2)
print("Time:",time.time()-s)
```

Input

```
10000
```

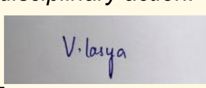
Output

```
Time: 0.0
```

Run Save Submit

Code of Conduct

I ______V.LASYA(192524186)____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _

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RUBRICS FOR CALCULATION

Experiments					
Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understand ing of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimen tal Design	30	Well-planned, systematic implementation with optimal solution	Correct implementatio n with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementatio n

Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Marks Evaluation:

Experiments	Understanding of Concepts **(25)**	Experimental Design **(30)**	Use of Tools **(20)**	Analysis of Results **(15)**	Documentation **(10)**	**Total Marks (Each - 50)**
Ex. 1						
Ex. 2						
					Overall Total (100)	**100 %**

Faculty In-charge	**Course Coordinator**	**Head of Department**
Name: ___________________ Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________